Advanced Plane Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In triangle ABC, the angle bisector from A meets side BC at point D, dividing it such that $BD = 6$ and $DC = 9$. If $AB = 8$, find AC, using the angle bisector theorem ($\frac{BD}{DC} = \frac{AB}{AC}$).

$\frac{6}{9} = \frac{8}{AC}$, so $6 \cdot AC = 72$, giving $AC = 12$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the exact area of a regular hexagon with side length 4.

Step 1: A regular hexagon is partitioned into 6 congruent equilateral triangles of side length $s = 4$. Step 2: Area of one equilateral triangle: $A_1 = \frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}(4^2) = 4\sqrt{3}$ Step 3: Total area of hexagon: $A = 6 \times 4\sqrt{3} = \mathbf{24\sqrt{3}} \approx 41.57$

3 **[Foundational]** • **Target: Circle theorems: tangent-chord and secant angles**

Two secants are drawn from an external point to a circle, intercepting arcs of 100° and 30°. Find the measure of the angle formed at the external point, using the formula (difference of arcs) ÷ 2.

Angle $= \frac{100^\circ - 30^\circ}{2} = \frac{70^\circ}{2} = 35^\circ$.

4 **[Foundational]** • **Target: Circle theorems: tangent-chord and secant angles**

Two chords intersect inside a circle, intercepting arcs of 80° and 40°. Find the measure of the angle formed at the intersection, using the formula (sum of arcs) ÷ 2.

Angle $= \frac{80^\circ + 40^\circ}{2} = \frac{120^\circ}{2} = 60^\circ$.

5 **[Foundational]** • **Target: Power of a point (secant and tangent segment relationships)**

A tangent segment from an external point has length 8, and a secant from the same point passes through the circle with external segment length 4. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

$8^2 = 4 \times (\text{whole secant})$, so $64 = 4w$, giving $w = 16$.

6 **[Foundational]** • **Target: Area of regular polygons**

Find the area of a regular pentagon with side length 6 and apothem 4.13, using $A = \frac{1}{2} \times \text{perimeter} \times \text{apothem}$.

Perimeter $= 5(6) = 30$. Area $= \frac{1}{2}(30)(4.13) = 61.95$.

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7 [Foundational] • Target: Similar solids and scale factor in 3D

Two similar cylinders have a scale factor of $\frac{2}{3}$. If the smaller cylinder has volume 40, find the volume of the larger cylinder.

Volumes of similar solids scale by the CUBE of the linear scale factor: $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$. Larger volume: $40 \times \frac{27}{8} = 135$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Composite 3D solids

A solid consists of a cylinder of radius 3 and height 8, topped with a hemisphere of the same radius. Find the total volume, in terms of π.

Cylinder volume: $\pi r^2 h = \pi(9)(8) = 72\pi$. Hemisphere volume: $\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(27) = 18\pi$. Total volume: $72\pi + 18\pi = 90\pi$.

9 [Intermediate] • Target: Triangle centers: centroid and circumcenter

Find the centroid of the triangle with vertices $(2, 3)$, $(8, 3)$, $(5, 9)$, using the formula (average of the vertex coordinates).

Centroid $= \left(\frac{2+8+5}{3}, \frac{3+3+9}{3}\right) = \left(\frac{15}{3}, \frac{15}{3}\right) = (5, 5)$.

10 [Intermediate] • Target: Heron's formula

A triangle has sides 7, 8, and 9. Find its area using Heron's formula, $A = \sqrt{s(s-a)(s-b)(s-c)}$ where s is the semi-perimeter.

$s = \frac{7+8+9}{2} = 12$. Area $= \sqrt{12(12-7)(12-8)(12-9)} = \sqrt{12(5)(4)(3)} = \sqrt{720} = 12\sqrt{5}$.

11 [Intermediate] • Target: A multi-part circle geometry problem

This question has two parts. A circle has radius 10. A chord is drawn 6 units from the center. (a) Find the length of the chord, using the relationship between the radius, the distance from center to chord, and half the chord length (a right triangle relationship). (b) Find the area of the minor segment cut off by this chord, to the nearest whole number, given that the central angle subtended by the chord is approximately 106.3° and $\sin(106.3^\circ) \approx 0.960$.

(a) Half the chord length: $\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64} = 8$. Full chord length: 16. (b) Segment area = sector area – triangle area. Sector area: $\frac{106.3}{360} \times \pi(10)^2 \approx 0.2953 \times 314.16 \approx 92.8$. Triangle area: $\frac{1}{2}(10)(10)\sin(106.3^\circ) \approx 50(0.960) = 48$. Segment area: $92.8 - 48 \approx 45$.

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12 **[Intermediate]** • **Target: Angle bisectors and the angle bisector theorem**

In triangle ABC, the angle bisector from A meets side BC at point D, dividing it such that $BD = 6$ and $DC = 9$. If $AB = 8$, find AC, using the angle bisector theorem ($\frac{BD}{DC} = \frac{AB}{AC}$).

$\frac{6}{9} = \frac{8}{AC}$, so $6 \cdot AC = 72$, giving $AC = 12$.

13 **[Intermediate]** • **Target: The 30-60-90 and 45-45-90 triangles in composite figures**

An equilateral triangle has side length 10. Find its height and area, using the fact that the height splits it into two 30-60-90 triangles.

The height corresponds to the longer leg of a 30-60-90 triangle with hypotenuse 10 and shorter leg 5: height $= 5\sqrt{3}$. Area $= \frac{1}{2}(10)(5\sqrt{3}) = 25\sqrt{3}$.

14 **[Intermediate]** • **Target: Circumscribed and inscribed circles**

A square is inscribed in a circle of radius 5. Find the side length of the square, using the fact that the diagonal of the square equals the diameter of the circle.

Diameter $= 2(5) = 10$. For a square, diagonal $= s\sqrt{2}$, so $s\sqrt{2} = 10$, giving $s = \frac{10}{\sqrt{2}} = 5\sqrt{2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Regular polygon interior and exterior angles** • **Cross-Lesson: 8.01 Statistics, Counting & Probability**

Find the measure of each interior angle and each exterior angle of a regular octagon.

Sum of interior angles of an n-gon: $(n-2)180^\circ$. For $n = 8$: $(6)(180^\circ) = 1080^\circ$. Each interior angle: $\frac{1080^\circ}{8} = 135^\circ$. Each exterior angle (supplementary to the interior angle): $180^\circ - 135^\circ = 45^\circ$.

16 **[Challenging]** • **Target: A multi-part composite figure and circle problem** • **Cross-Lesson: 1.01 Advanced Number Theory**

This question has three parts. A running track consists of a rectangle with two semicircular ends. The rectangle is 80 meters long and 40 meters wide, and the semicircles are attached to the two 40-meter-wide ends. (a) Find the radius of each semicircle. (b) Find the total perimeter (distance around) the track, in terms of π. (c) Find the total enclosed area of the track shape, in terms of π.

(a) The semicircles are attached to the 40-meter ends, so the diameter of each semicircle is 40, giving radius $= 20$. (b) Perimeter $= 2(\text{rectangle length}) + 2(\text{semicircle arc length})$. The two semicircles together form one full circle's circumference: $2\pi(20) = 40\pi$. Total perimeter: $2(80) + 40\pi = 160 + 40\pi$ meters. (c) Area $= \text{rectangle area} + \text{combined semicircle area}$. Rectangle area: $80 \times 40 = 3200$. The two semicircles together form one full circle: $\pi(20)^2 = 400\pi$. Total area: $3200 + 400\pi$ square meters.

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17 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Advanced Algebra & Systems

A tangent segment from an external point has length 8, and a secant from the same point passes through the circle with external segment length 4. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

$8^2 = 4 \times (\text{whole secant})$, so $64 = 4w$, giving $w = 16$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Advanced Functions

Find the area of a regular pentagon with side length 6 and apothem 4.13, using $A = \frac{1}{2} \times \text{perimeter} \times \text{apothem}$.

Perimeter $= 5(6) = 30$. Area $= \frac{1}{2}(30)(4.13) = 61.95$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Advanced Trigonometry

Two similar cylinders have a scale factor of $\frac{2}{3}$. If the smaller cylinder has volume 40, find the volume of the larger cylinder.

Volumes of similar solids scale by the CUBE of the linear scale factor: $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$. Larger volume: $40 \times \frac{27}{8} = 135$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the measure of each exterior angle of a regular hexagon.

The sum of all exterior angles of any convex polygon is 360°. For a regular hexagon with $n = 6$ equal sides: $\text{Exterior angle} = \frac{360^\circ}{n} = \frac{360^\circ}{6} = \mathbf{60^\circ}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A tangent segment from an external point has length 30, and a secant from the same point passes through the circle with external segment length 20. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 25

b. 30

c. 45 ✓

d. 46

$30^2 = 20 \times (\text{whole secant})$, so $900 = 20w$, giving $w = 45$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A tangent segment from an external point has length 6, and a secant from the same point passes through the circle with external segment length 4. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 6

b. 10

c. 5

d. 9 ✓

$6^2 = 4 \times (\text{whole secant})$, so $36 = 4w$, giving $w = 9$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Two similar cylinders have a scale factor of $\frac{2}{3}$. If the smaller cylinder has volume 32, find the volume of the larger cylinder.

a. 110

b. 32

c. 108 ✓

d. 72

Volumes of similar solids scale by the CUBE of the linear scale factor: $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$. Larger volume: $32 \times \frac{27}{8} = 108$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC, the circumcenter lies on the midpoint of the longest side. What type of triangle is ΔABC?

a. Acute

b. **Right** ✓

c. Obtuse

d. Equilateral

By Thales's theorem, the circumcenter of a triangle lies on a side if and only if the triangle is a right triangle, where the hypotenuse is the diameter of the circumcircle.

25 [Foundational] • Target: Composite 3D solids

Two chords intersect inside a circle, intercepting arcs of 170° and 76°. Find the measure of the angle formed at the intersection, using the formula (sum of arcs) ÷ 2.

a. 246°

b. 133°

c. 47°

d. 123° ✓

Angle $= \frac{170^\circ + 76^\circ}{2} = \frac{246^\circ}{2} = 123^\circ$.

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26 [Foundational] • Target: Triangle centers: centroid and circumcenter

Two secants are drawn from an external point to a circle, intercepting arcs of 157° and 43°. Find the measure of the angle formed at the external point, using the formula (difference of arcs) ÷ 2.

a. 67°

b. 114°

c. 100°

d. 57° ✓

Angle $= \frac{157^\circ - 43^\circ}{2} = \frac{114^\circ}{2} = 57^\circ$.

27 [Foundational] • Target: Heron's formula

Two chords intersect inside a circle, intercepting arcs of 117° and 17°. Find the measure of the angle formed at the intersection, using the formula (sum of arcs) ÷ 2.

a. 134°

b. 50°

c. 77°

d. 67° ✓

Angle $= \frac{117^\circ + 17^\circ}{2} = \frac{134^\circ}{2} = 67^\circ$.

28 [Foundational] • Target: A multi-part circle geometry problem

Two chords intersect inside a circle, intercepting arcs of 12° and 104°. Find the measure of the angle formed at the intersection, using the formula (sum of arcs) ÷ 2.

a. 68°

b. 58° ✓

c. 46°

d. 116°

Angle $= \frac{12^\circ + 104^\circ}{2} = \frac{116^\circ}{2} = 58^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Angle bisectors and the angle bisector theorem

Two secants are drawn from an external point to a circle, intercepting arcs of 184° and 14°. Find the measure of the angle formed at the external point, using the formula (difference of arcs) ÷ 2.

a. 99°

b. 170°

c. 85° ✓

d. 95°

Angle $= \frac{184^\circ - 14^\circ}{2} = \frac{170^\circ}{2} = 85^\circ$.

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30 [Intermediate] • Target: The 30-60-90 and 45-45-90 triangles in composite figures

Two secants are drawn from an external point to a circle, intercepting arcs of 118° and 52°. Find the measure of the angle formed at the external point, using the formula (difference of arcs) ÷ 2.

a. 43°

b. 33° ✓

c. 85°

d. 66°

Angle $= \frac{118^\circ - 52^\circ}{2} = \frac{66^\circ}{2} = 33^\circ$.

31 [Intermediate] • Target: Circumscribed and inscribed circles

A tangent segment from an external point has length 30, and a secant from the same point passes through the circle with external segment length 20. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 25

b. 30

c. 45 ✓

d. 46

$30^2 = 20 \times (\text{whole secant})$, so $900 = 20w$, giving $w = 45$.

32 [Intermediate] • Target: Regular polygon interior and exterior angles

A tangent segment from an external point has length 28, and a secant from the same point passes through the circle with external segment length 16. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 33

b. 49 ✓

c. 28

d. 50

$28^2 = 16 \times (\text{whole secant})$, so $784 = 16w$, giving $w = 49$.

33 [Intermediate] • Target: A multi-part composite figure and circle problem

A tangent segment from an external point has length 22, and a secant from the same point passes through the circle with external segment length 11. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 45

b. 44 ✓

c. 33

d. 22

$22^2 = 11 \times (\text{whole secant})$, so $484 = 11w$, giving $w = 44$.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A tangent segment from an external point has length 8, and a secant from the same point passes through the circle with external segment length 2. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 32 ✓

b. 30

c. 33

d. 8

$8^2 = 2 \times (\text{whole secant})$, so $64 = 2w$, giving $w = 32$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A tangent segment from an external point has length 12, and a secant from the same point passes through the circle with external segment length 4. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 12

b. 37

c. 36 ✓

d. 32

$12^2 = 4 \times (\text{whole secant})$, so $144 = 4w$, giving $w = 36$.

36 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A tangent segment from an external point has length 12, and a secant from the same point passes through the circle with external segment length 9. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 16 ✓

b. 12

c. 7

d. 17

$12^2 = 9 \times (\text{whole secant})$, so $144 = 9w$, giving $w = 16$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A triangle has sides 6, 8, and 10. Find its area using Heron's formula, $A = \sqrt{s(s-a)(s-b)(s-c)}$ where s is the semi-perimeter.

a. 26

b. 27

c. 24 ✓

d. 22

$s = \frac{6+8+10}{2} = 12$. Area $= \sqrt{12(6)(4)(2)} = \sqrt{576} = 24$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Advanced Functions

A triangle has sides 10, 10, and 16. Find its area using Heron's formula, $A = \sqrt{s(s-a)(s-b)(s-c)}$ where s is the semi-perimeter.

a. 46

b. 36

c. 48 ✓

d. 50

$s = \frac{10+10+16}{2} = 18$. Area $= \sqrt{18(8)(8)(2)} = \sqrt{2304} = 48$.

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Advanced Trigonometry

A solid consists of a cylinder of radius 6 and height 11, topped with a hemisphere of the same radius. Find the total volume, in terms of π.

a. 540π ✓

b. 396π

c. 546π

d. 144π

Cylinder volume: $\pi r^2 h = \pi(36)(11) = 396\pi$. Hemisphere volume: $\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(216) = 144\pi$. Total volume: $396\pi + 144\pi = 540\pi$.

40 [Challenging] • Target: Circle theorems: tangent-chord and secant angles • Cross-Lesson: 5.01 Pre-Calculus Concepts

A solid consists of a cylinder of radius 3 and height 5, topped with a hemisphere of the same radius. Find the total volume, in terms of π.

a. 66π

b. 18π

c. 45π

d. 63π ✓

Cylinder volume: $\pi r^2 h = \pi(9)(5) = 45\pi$. Hemisphere volume: $\frac{2}{3}\pi r^3 = \frac{2}{3}\pi(27) = 18\pi$. Total volume: $45\pi + 18\pi = 63\pi$.

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41 [Challenging] • Target: Power of a point (secant and tangent segment relationships) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Two similar cylinders have a scale factor of $\frac{2}{3}$. If the smaller cylinder has volume 32, find the volume of the larger cylinder.

a. 110

b. 32

c. 108 ✓

d. 72

Volumes of similar solids scale by the CUBE of the linear scale factor: $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$. Larger volume: $32 \times \frac{27}{8} = 108$.

42 [Challenging] • Target: Area of regular polygons • Cross-Lesson: 8.01 Statistics, Counting & Probability

Two similar cylinders have a scale factor of $\frac{3}{5}$. If the smaller cylinder has volume 162, find the volume of the larger cylinder.

a. 753

b. 450

c. 162

d. 750 ✓

Volumes of similar solids scale by the CUBE of the linear scale factor: $\left(\frac{5}{3}\right)^3 = \frac{125}{27}$. Larger volume: $162 \times \frac{125}{27} = 750$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Advanced Number Theory

A tangent segment from an external point has length 22, and a secant from the same point passes through the circle with external segment length 11. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 45

b. 44 ✓

c. 33

d. 22

$22^2 = 11 \times (\text{whole secant})$, so $484 = 11w$, giving $w = 44$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Advanced Algebra & Systems

A tangent segment from an external point has length 15, and a secant from the same point passes through the circle with external segment length 5. Find the length of the entire secant segment, using $(\text{tangent})^2 = (\text{external secant})(\text{whole secant})$.

a. 40

b. 45 ✓

c. 15

d. 46

$15^2 = 5 \times (\text{whole secant})$, so $225 = 5w$, giving $w = 45$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Advanced Functions

In ΔABC, medians intersect at the centroid G. If median AM has length 18 cm, what is the length of AG?

a. 6 cm

b. 9 cm

c. 12 **cm** ✓

d. 15 cm

The centroid divides each median in a $2:1$ ratio from vertex to midpoint: $AG = \frac{2}{3}AM = \frac{2}{3}(18) = 12$ cm.

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